Java Assignment - 3

Name: Saloman Raju Sarikokkula

Overview of the week

S.No	Date	Topic
1.	23/03/2026	Looping Statements
2.	25/03/2026	Arrays
3.	26/03/2026	Strings

Looping Statements

- Used to execute a block of code multiple times.
- **Three** types:
    1. `for` loop
    2. `while` loop
    3. `do-while` loop

1. `for()` loop

Best when you the know exact count of iterations

Syntax:

```
for(initialization; condition; updatation){
    stmts..
}
```

Example:

```
for (int i = 0; i < 5; i++) {
    System.out.println("Count: " + i);
}
```

Enhanced`for()` loop

Designed for iterating arrays and collections, no index needed.

Syntax:

```
for(<DataType> var: <array/collection>){
    stmts..
}
```

Example:

```
int[] numbers = {1, 2, 3, 4, 5};
for (int num : numbers) {
    System.out.println(num);
}
```

2. `while()` loop

Runs as long as the condition is true, best when the exact count of iterations is unknown.

Syntax:

```
while(condition){
    stmts..
}
```

Example:

```
int i = 0;
while (i < 5) {
    System.out.println("Count: " + i);
     i++
}
```

3. `do-while()` loop

Runs as long as the condition is true, but executes at least once unlike while loop.

Syntax:

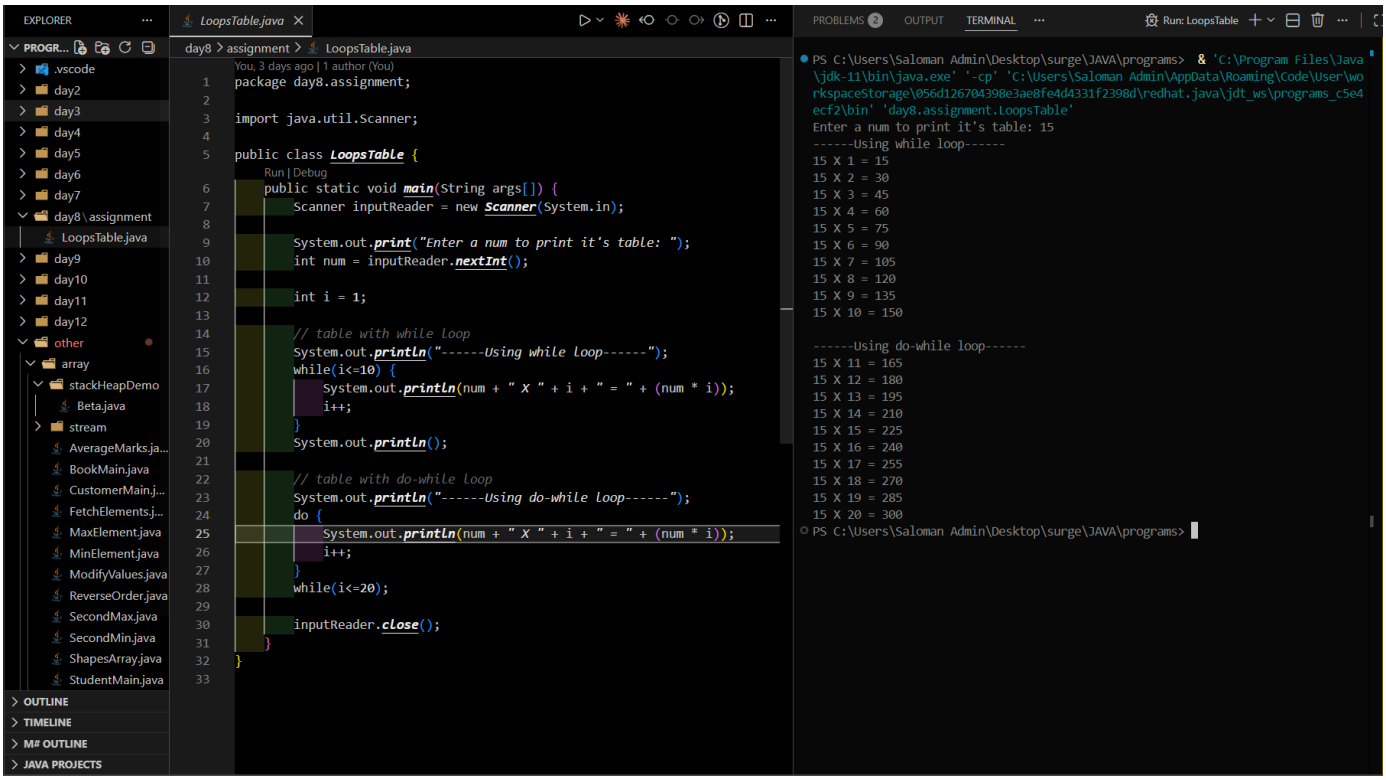
```
do {
    stmts..
} while(condition);
```

Example:

```java
int i = 0;
do {
    System.out.println("Count: " + i);
    i++;
} while (i < 5);
```

Execution on VS Code

Program to print table using while and do-while loops with scanner class.



Arrays

- A container with fixed size which stores the ordered collection of homogeneous data.
- A non-primitive data which is developed to tackle the single value storing prob of primitive data types.
- **Two** types:
    1. Single Dimensional - Data is stored in a single dimensional contiguous memory.
    2. Multi-Dimensional - It's like an `Array of Arrays` - like table or grid.
        1. Square [rows = cols]
        2. Jagged [each row has its own length]

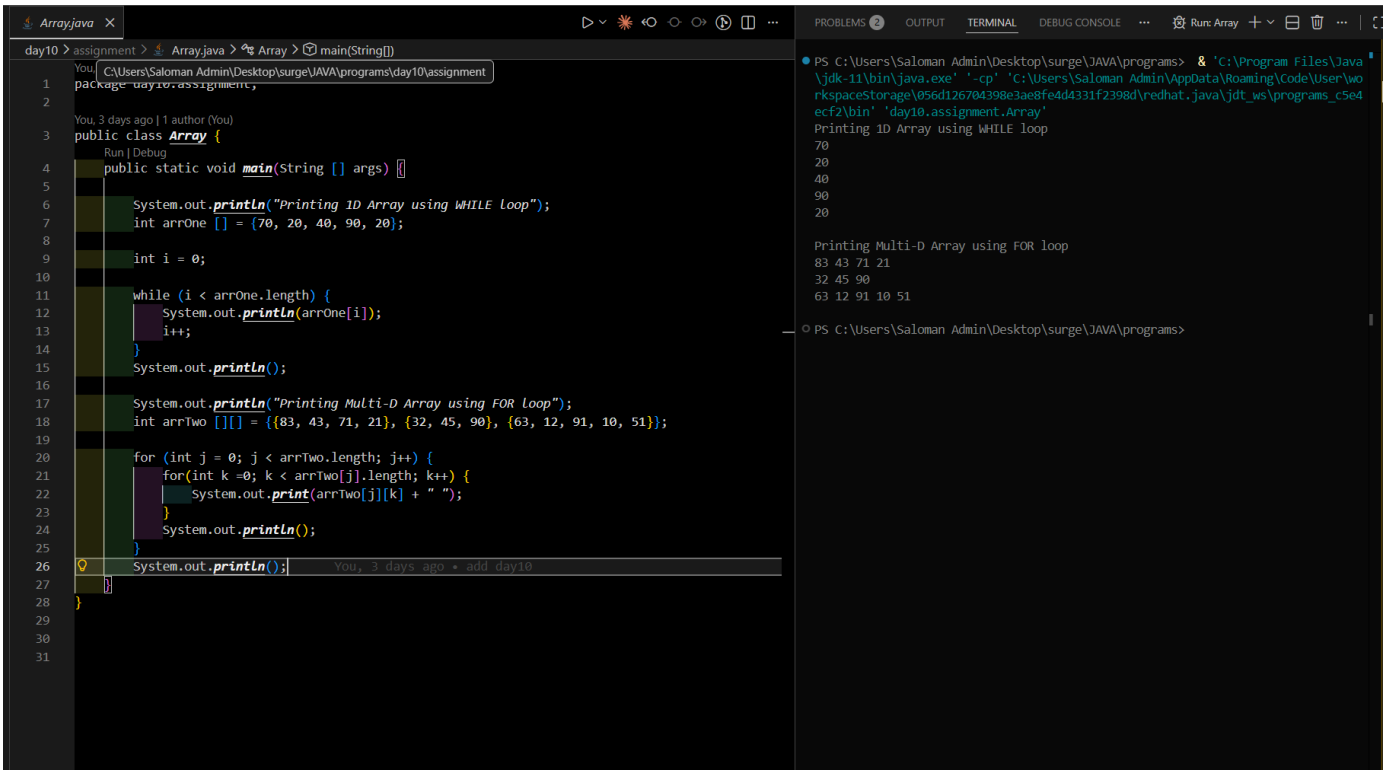
Syntax with example:

```java
// Declaration
int[] nums;       // 1D
int [] [] nums    // 2D

// Declaration + allocation
int[] nums = new int[5];  // 5 elements, all default to 0        //1D
int[][] nums = new int[3][3];  // 9 elements, all default to 0   //2D

// Declaration + initialization
int[] nums = {10, 20, 30, 40, 50};                              //1D
int[][] nums = {{10, 20, 30}, {40, 50, 60}, {70, 80, 90}};      //21D

// Explicit form
int[] nums = new int[]{10, 20, 30, 40, 50};                              //1D
int[][] nums = new int[][]{{10, 20, 30}, {40, 50, 60}, {70, 80, 90}}; //2D
```

Execution on VS Code

Program to print the:

- 1D array with while loop
- Multi D array with for loop



Strings

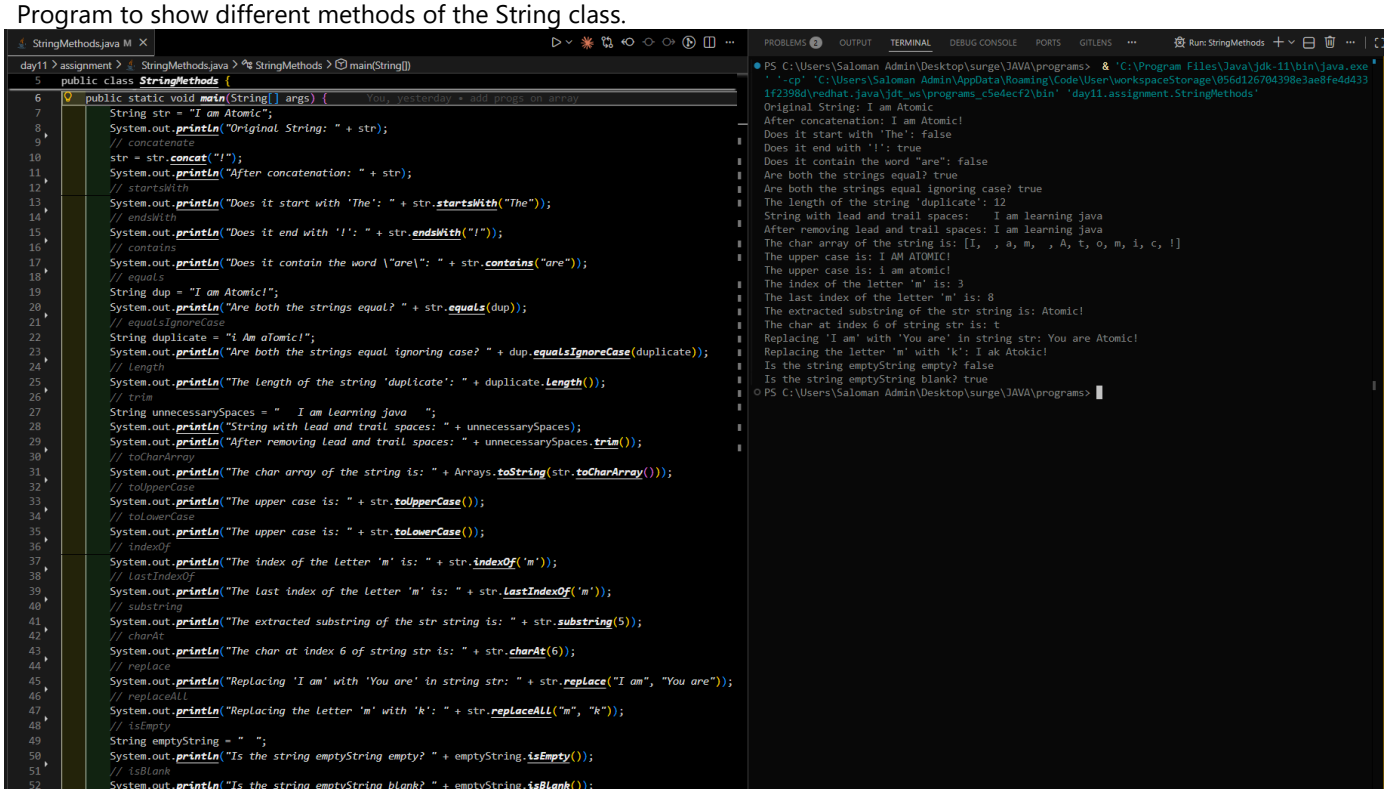
- It's a pre-defined class from `java.util` package which stores the sequence of characters.
- It's a non-primitive data type whose data is stored in the String pool in Heap.
- They are immutable in terms of address.

Syntax with example:

```
// String literal (stored in String Pool)
String name = "Hello";

// Using new keyword (stored in Heap)
String name = new String("Hello");
```

String methods:

```
String s = "Hello, World!";

s.length()                    // 13
s.charAt(0)                   // 'H'
s.indexOf("World")            // 7
s.substring(7)                // "World!"
s.substring(7, 12)            // "World"
s.toUpperCase()               // "HELLO, WORLD!"
s.toLowerCase()               // "hello, world!"
s.trim()                      // removes leading/trailing spaces
s.replace("World", "Java")  // "Hello, Java!"
s.contains("World")           // true
s.startsWith("Hello")         // true
s.endsWith("!")               // true
s.isEmpty()                   // false
s.split(", ")                 // ["Hello", "World!"]
```

Execution on VS Code

Program to show different methods of the String class.